

LAB REPORT

11th May 2025



Network Forensics

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For the class of Computer Forensic, taught by

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1 Introduction

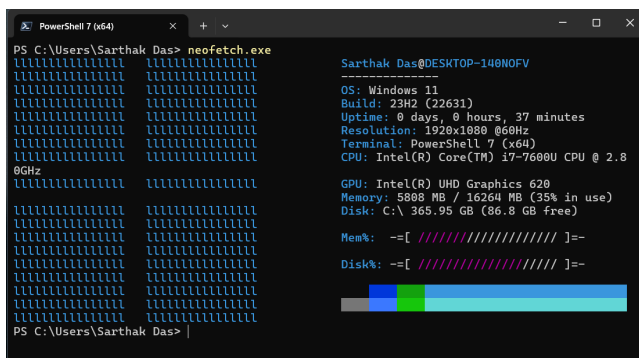
1.1 Objectives

This report is a documentation of my lab work on network forensics. The primary objectives of this lab are two-fold:

- To investigate web attacks using Splunk.
- To analyze network dump files using Wireshark.

1.2 System Information

For the purposes of this lab, a Lenovo ThinkPad X270 was used, having the following system specifications: Intel Core i7-7600U CPU @ 2.80GHz with Intel UHD Graphics 620, 16GB RAM, 1TB SSD running operating system Windows 11 23H2 (Build 22631).



```
PS C:\Users\Sarthak Das> neofetch.exe

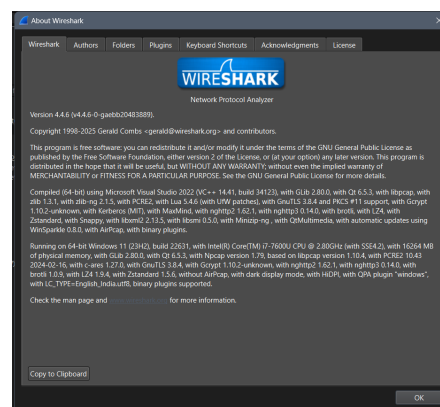
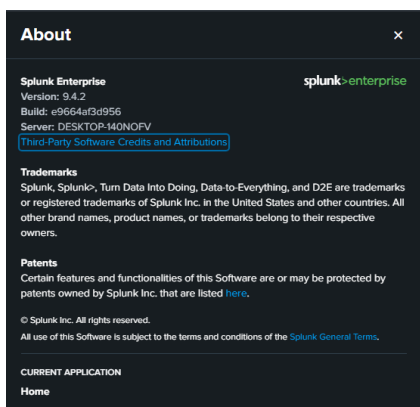
Sarthak Das@DESKTOP-148NOFV
-----
OS: Windows 11
Build: 23H2 (22631)
Uptime: 0 days, 0 hours, 37 minutes
Resolution: 1920x1080 @60Hz
Terminal: PowerShell 7 (x64)
CPU: Intel(R) Core(TM) i7-7600U CPU @ 2.8
    GHz
GPU: Intel(R) UHD Graphics 620
Memory: 5808 MB / 16264 MB (35% in use)
Disk: C:\ 365.95 GB (36.8 GB free)

Mem%:  --[ ##### ]--
Disk%: --[ ##### ]--

PS C:\Users\Sarthak Das> |
```

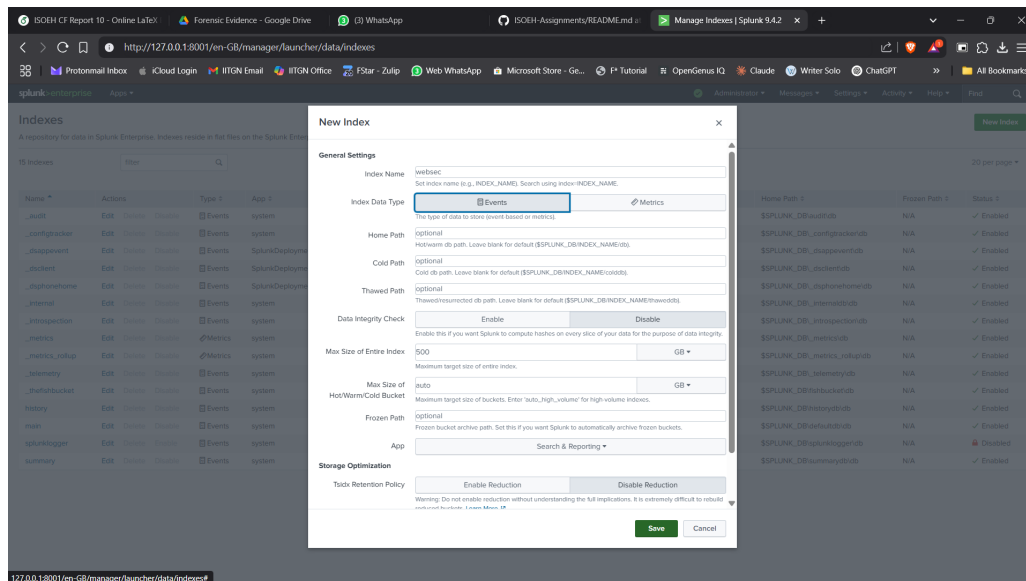
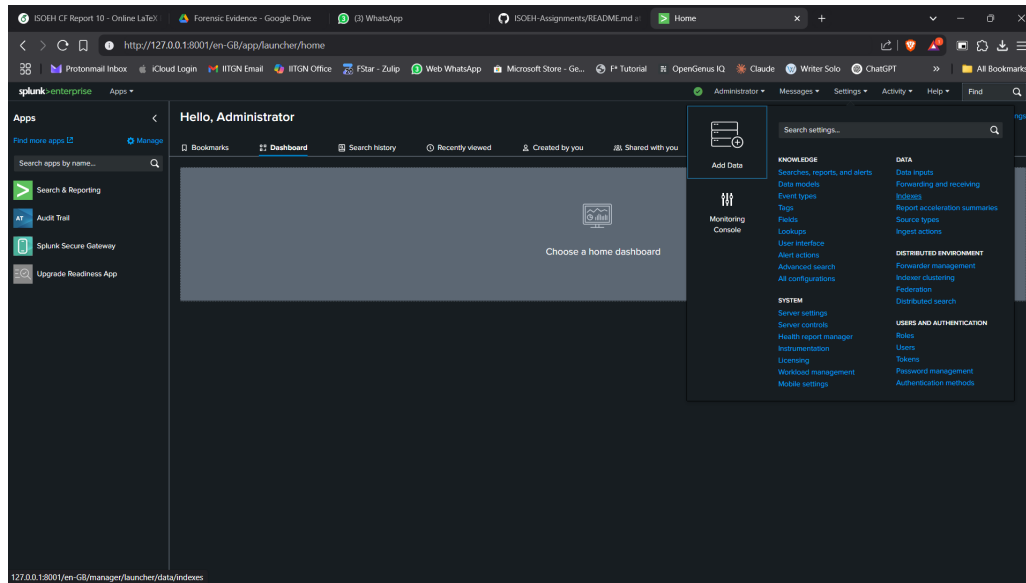
1.3 Software Used

The software used for this practical are Splunk Enterprise Version 9.4.2 Build e9664af3d956 and Wireshark Version 4.4.6.

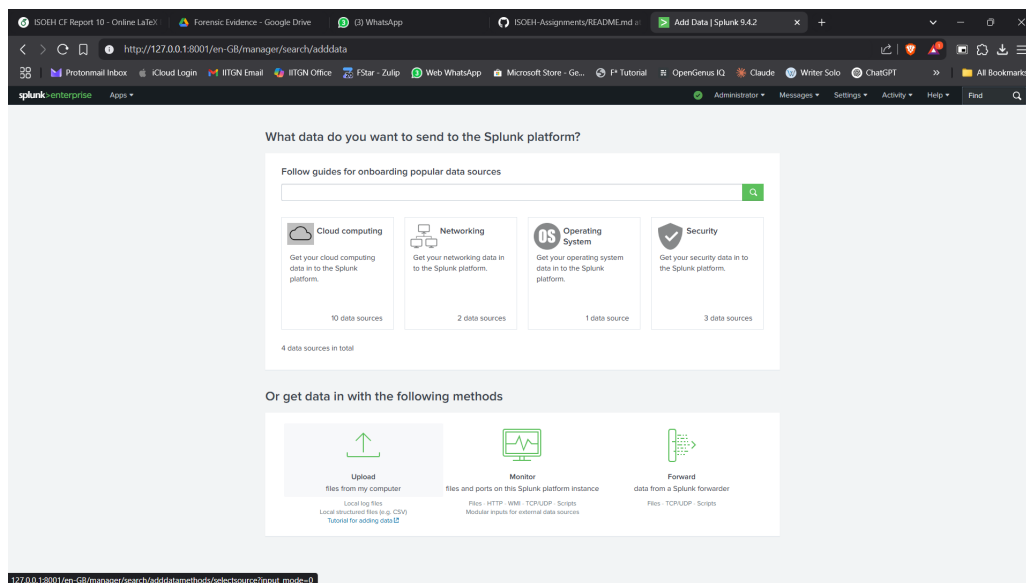
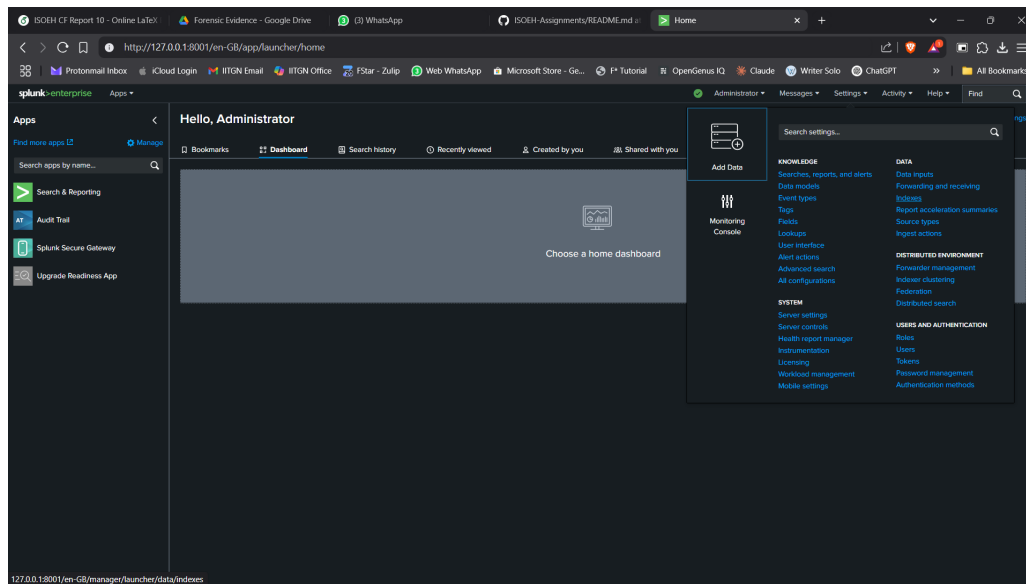


2 Objective 1: Web Attack Investigation

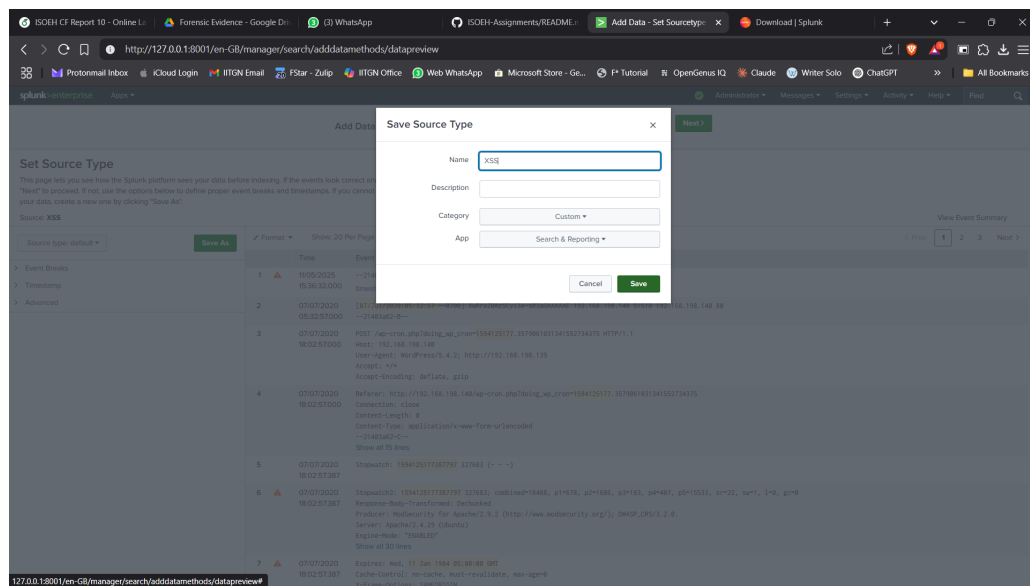
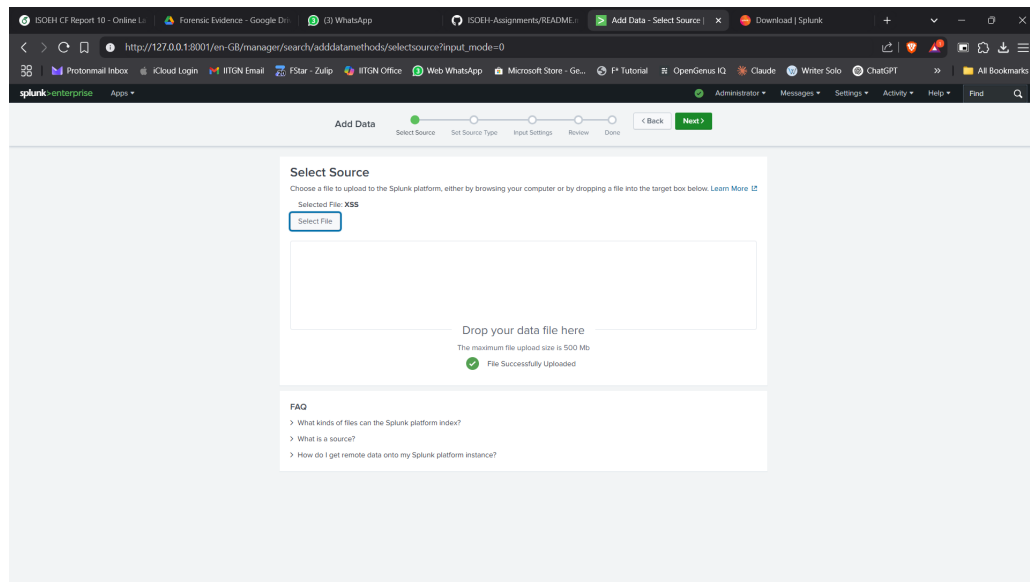
We begin by opening Splunk Enterprise, logging in, going to **Settings** > *Indexes* and creating a new index by the name of **websec**.

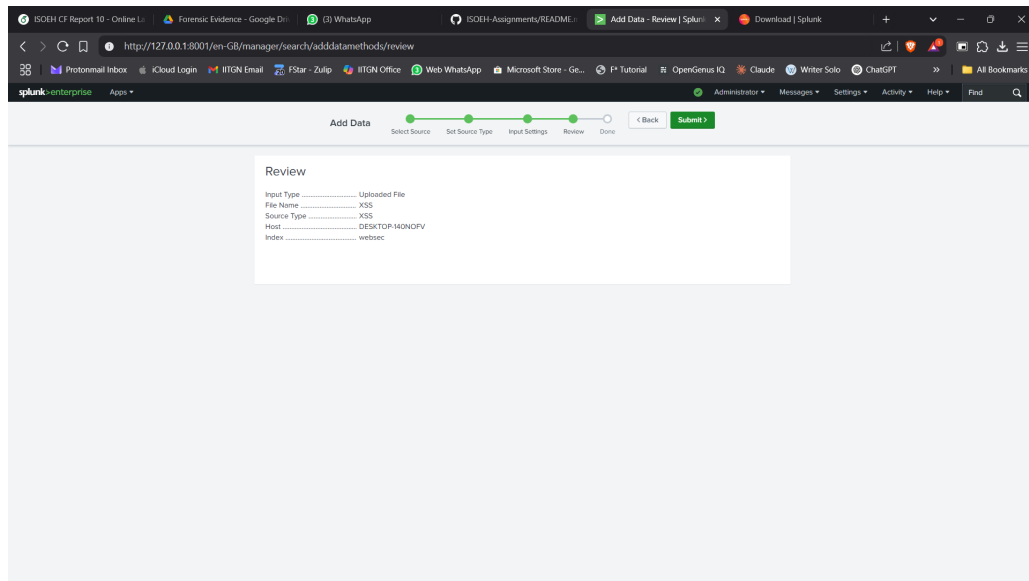
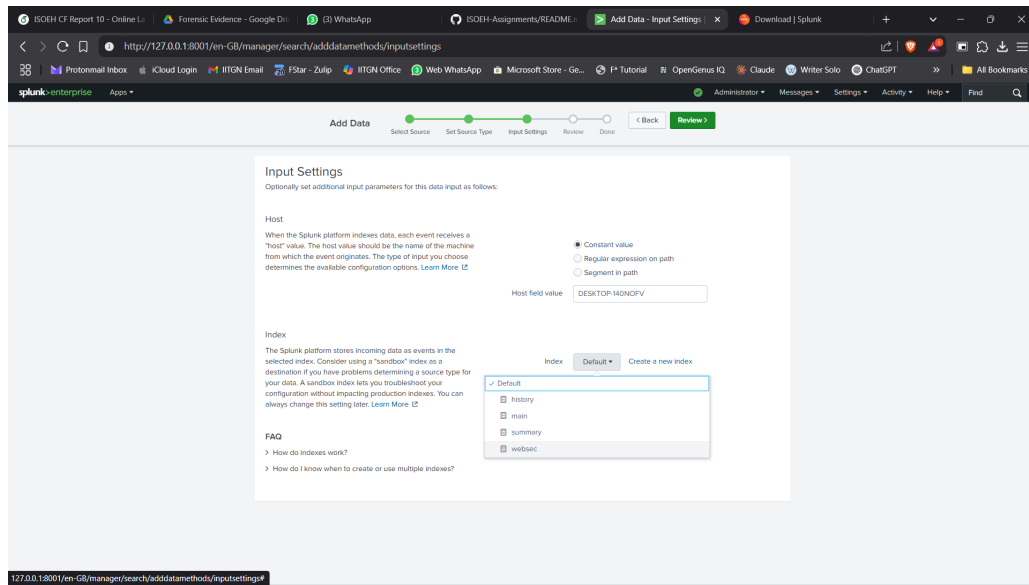


Once the index is created, we need to upload our data. For the purposes of this lab, we were provided with three log files – *Brute-Force.txt*, *SQL Injection.log* and *XSS*. This is done by going to **Settings > Add Data**. On the console, we select **Upload files from my computer**.

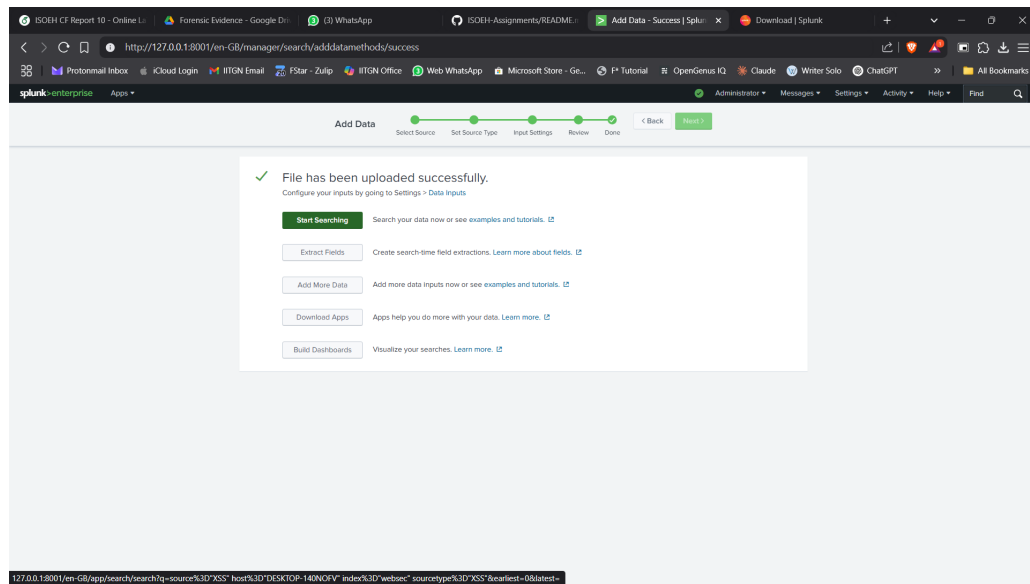


For each file, we upload it, name the source type accordingly and set the index to `websec`.

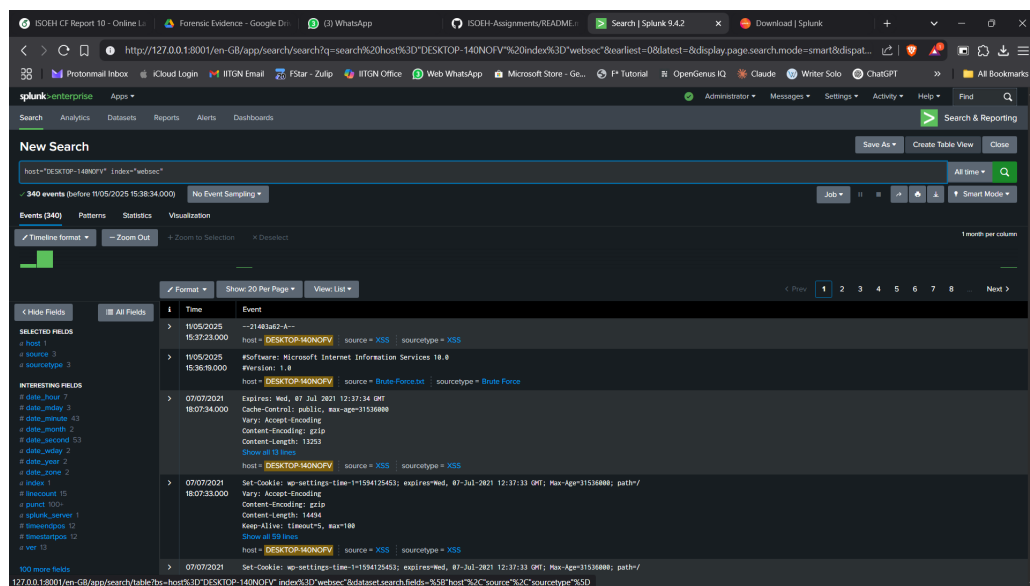




Once all files have been uploaded, we can click on **Start Searching** to begin our analysis.

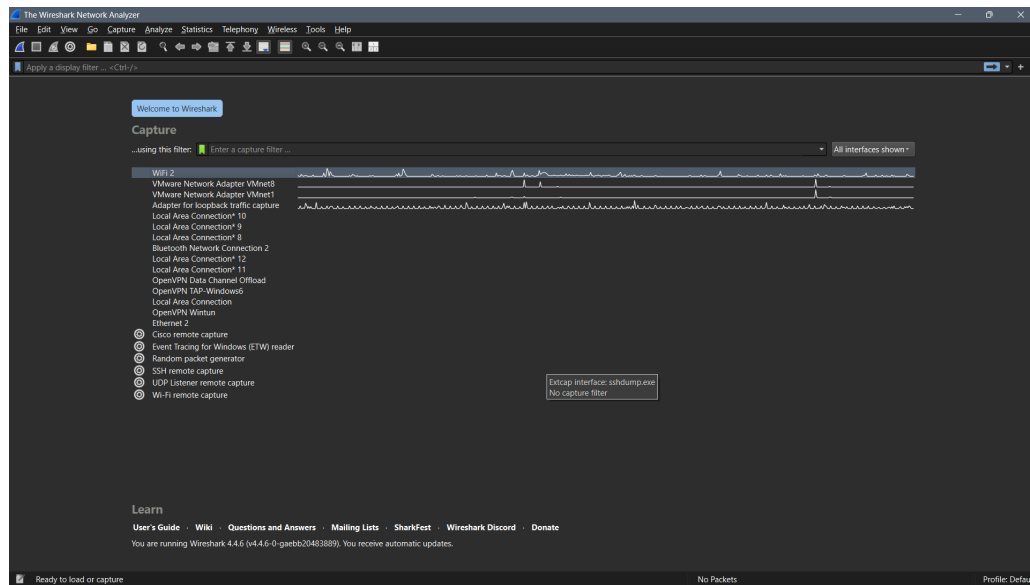


We can now tune the search query to our liking, and analyze the logs in raw or table view.

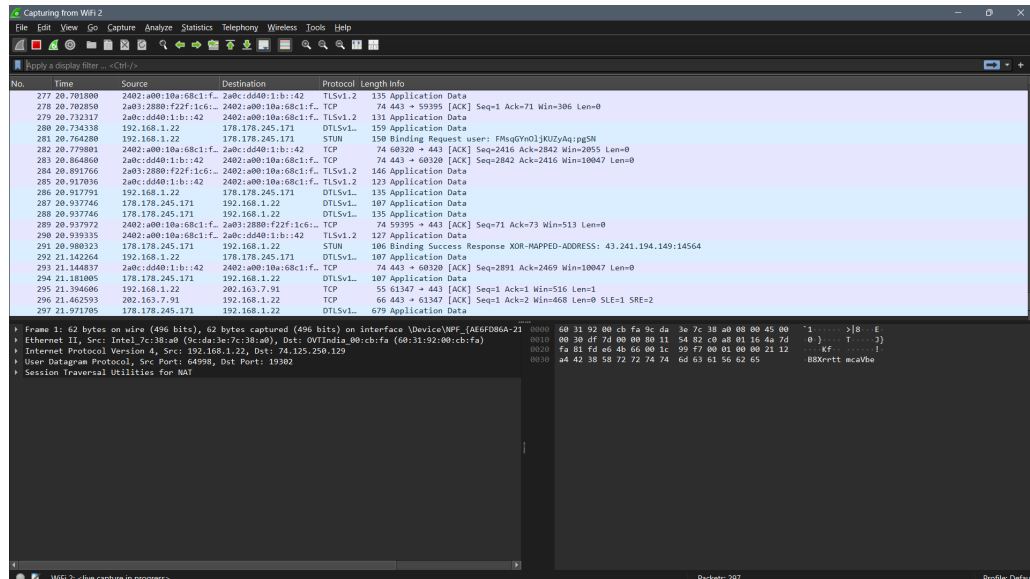


3 Objective 2: Network Dump Analysis

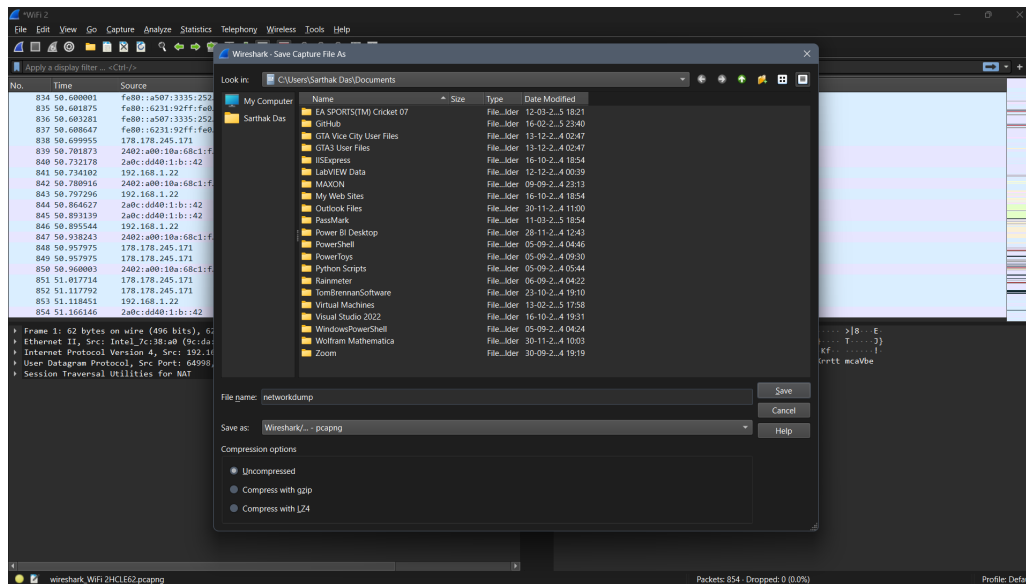
Before we begin, we must create a network dump file to analyze. This can be done in Wireshark itself. First, we open Wireshark and select the network interface to capture data packets from.



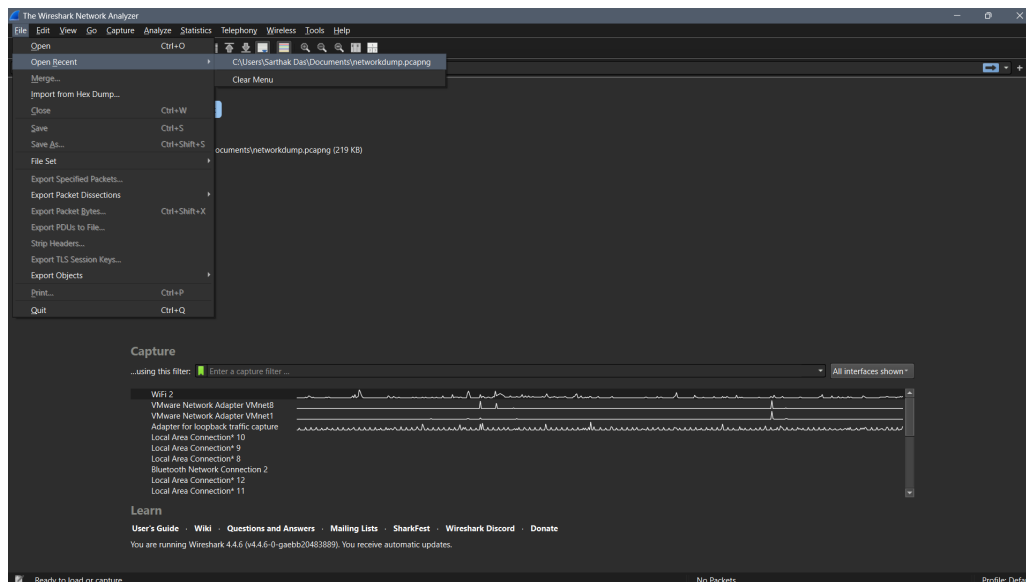
Wireshark starts capturing data packets for us from the WiFi 2 interface in our case.



We can stop the capturing after some time, and save the capture file with a .pcapng extension.



This is our network dump file. We can analyze this file in Wireshark itself. In order to do so, we select **Files > Open** and browse to our file. Alternatively, we can select *Open Recent* and choose our file from there, since we used Wireshark to create this file.



We can now use Wireshark's inbuilt analysis tools to analyze the data packets.

